Tehreem Kamal

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Education

Sept/2025 – Continue
MS Artificial Intelligence & Big Data Analytics Novosibirsk State University, Russia
Current CGPA: 5/5

Nov/2019 – Nov/2022
B.E Computer System Engineering Dawood University of Engineering and Technology, Pakistan
Thesis Title: *Automated Industrial Inspection System for Plastic Bottle Defects*
CGPA: 3.35/4.0

Research Interest

- **Machine Learning**
- **Computer Vision**
- **medical imaging**
- **Deep Learning**
- **Software Development**

Projects

2022
Automated Industrial Inspection System for Plastic Bottle Defects Final Year Project
Developed an AI-based automated inspection system using deep learning (TensorFlow Lite) for real-time dent and crack detection in plastic bottles, integrated with a custom experimental testbed and servo motor-controlled imaging setup.

Employment History

June/2026 – Present
AI Researcher, Stream Data Analytics & Machine Learning Lab, Novosibirsk State University.

- Working on deep learning methods for white matter and gray matter segmentation in brain MRI of multiple sclerosis patients.
- Developing and validating medical image segmentation pipelines using PyTorch, nnU-Net, and MONAI, with a focus on handling MS lesions and anatomical variations.
- Performing MRI preprocessing, image normalization, and data augmentation for reliable brain tissue segmentation.
- Evaluating segmentation performance using Dice score, Hausdorff distance, and related medical image analysis metrics.
- Collaborating on MRI annotation, result validation, and interpretation for AI-based medical imaging research.

Employment History (continued)

May/2023 – Present

SOFTWARE DEVELOPER MINIME-LABS.

- Transforming design prototypes into responsive, interactive web applications, implementing dynamic front-end architectures, and seamlessly integrating with scalable, secure backend systems and relational databases.
- Currently working on the frontend of two projects: Ezehome and Sentry.

Nov/2022 – May/2023

AI ENGINEER NEURCOMPUTATIONAL LAB NCAI NED UNIVERSITY OF ENGINEERING AND TECHNOLOGY KARACHI.

- Developed Deep Learning model for defect detection in bottles using TensorFlow. Built and annotated a large dataset, performed hyperparameter tuning, and validated the model on extensive datasets to meet quality benchmarks. Developed a user-friendly web application using Django and Flask for seamless model deployment and interaction.

Jun/2022 – Jul/2022

UI/UX DEVELOPER INTERN NEURCOMPUTATIONAL LAB NCAI NED UNIVERSITY OF ENGINEERING AND TECHNOLOGY KARACHI.

- Developed advanced dashboards by applying wireframing, interactive prototyping, responsive UI design, and optimized UX flows.
- Utilized dynamic components, state management, data visualization tools, and user-centric design principles to create intuitive, high-performance interfaces.

Apr/2022 – Jun/2022

FRONT END DEVELOPER INTERN CROP2X PRIVATE LIMITED KARACHI, PAKISTAN.

- Developed responsive user interfaces from design prototypes and implemented seamless integration with backend services and relational databases to deliver fully functional, production-ready web applications.

Skills

Languages	Strong reading, writing, and speaking competencies in English, Urdu, Sindhi, and Punjabi.
Coding	Python, TensorFlow, PyTorch, SQL, LaTeX, sphinx,
Databases	MySQL and PostgreSQL .
Front end & Backend development	HTML, CSS, JavaScript, React.js, Flask, Django.
Data Analysis/ Machine Learning	Academic Research, Data Analysis, Machine Learning, Deep Learning, Natural Language Processing, and Computer Vision.

Professional & Academic Experience

Academic & Volunteer Activities

2019

TECHFEST'19, a one day seminar on Virtual Reality, Data Analytics and Smart Grid Haptics Technology By Sentec

Professional & Academic Experience (continued)

Awards & Achievements

2022 **National Technology Fund (Ignite)** Automated Industrial Inspection System for Plastic Bottle Defects Funded By Ignite.

MOOCs & Certifications

2023 **Artificial Intelligence Developer** by Presidential Initiative for Artificial Intelligence and Computing

2022 **Train YOLO V4 for Object Detection with Dataset** by Udemy

2020 **Web Development Bootcamp** by Udemy

2022 **Introduction to the critical Infrastructure Protection** by ICIP Opswate Academy

2020 **Intro to Data and Data Science** by 365 Datascience.

References

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MINIME-LABS Cyprus
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